

A Compact Eight-Element UWB MIMO Antenna for Radar Imaging and High-Speed Wireless Communications

Md. Abdul Awal[†], Md. Najmul Hossain^{†,*}, Sayed Shifat Ahmed[‡], and Tetsuya Shimamura[§]

[†]Department of Electrical, Electronic and Communication Engineering
Pabna University of Science and Technology, Pabna 6600, Bangladesh

[‡]Department of Electrical and Electronic Engineering
RTM Al-Kabir Technical University, Sylhet 3100, Bangladesh

[§]Graduate School of Science and Engineering, Saitama University
Saitama 338-8570, Japan

Email: awal.eece@gmail.com, najmul_eece@pust.ac.bd, shikor99@gmail.com, shima@mail.saitama-u.ac.jp

*Corresponding author

Abstract—This paper presents a compact and high-efficiency eight-element ultra-wideband (UWB) MIMO antenna for multi-gigabit wireless communication systems and UWB radar imaging applications. The proposed low-profile antenna has an overall size of $70 \times 38 \text{ mm}^2$, is implemented on an FR-4 substrate, and achieves an impedance bandwidth of 6.1 GHz. The antenna structure is optimized to minimize mutual coupling among the elements and maintain high isolation, enabling reliable spatial-diversity performance. The design exhibits a low envelope correlation coefficient ($\text{ECC} < 0.004$) across the operating band and a radiation efficiency of approximately 94%. Owing to its compact size, wide impedance bandwidth, stable gain, and robust radiation performance, the proposed MIMO antenna is a promising candidate for UWB radar imaging, through-wall sensing, and next-generation high-throughput short-range wireless communication systems.

Index Terms—Ultra-wideband (UWB), MIMO antenna, eight-element array, low-profile antenna, radar imaging, ECC.

I. INTRODUCTION

High quality of service and high data rates are fundamental requirements in modern wireless communication systems. With the extensive deployment of wireless networks, compact antennas with broadband operation and high data-transmission capability are increasingly required. Ultra-wideband (UWB) technology has attracted considerable attention for sensor imaging and short-range high-speed wireless communications [1], [2]. Due to efforts by the Telecommunications Industry Association and the Federal Communications Commission, a 7.5 GHz spectrum was allocated for UWB communication in 2002, accelerating its use in medical imaging, radar, and related applications [2]. Despite these merits, UWB systems are affected by multipath fading and low transmission power, which can reduce system performance and signal reliability [2], [3]. Multiple-input multiple-output (MIMO) technology helps overcome these limitations by improving channel capacity, throughput, and link reliability without requiring additional bandwidth or transmit power [3]. Spatial and polarization

diversity in compact MIMO antenna designs also mitigates fading and improves throughput in small-form-factor devices [4].

Several UWB MIMO antennas have been reported to improve bandwidth, isolation, and compactness. The design in [5] occupies $150 \times 70 \text{ mm}^2$ and operates over dual bands of 3.4–3.8 GHz and 10.5–14 GHz, with 9 GHz total bandwidth and a peak gain of 5.65 dB; however, its radiation efficiency is limited to 61%, and its ECC remains below 0.15. Rahayu *et al.* [6] presented a wideband antenna covering 2.75–12 GHz with 14.9 GHz bandwidth and a peak gain of 9.65 dB, but key MIMO metrics such as ECC, radiation efficiency, and physical dimensions were not fully reported. Govindan *et al.* [7] proposed a UWB MIMO antenna operating over 3.1–10.6 GHz with 3.4 dB gain, 89.3% efficiency, and $\text{ECC} < 0.18$. The antenna in [8] offers 2 GHz bandwidth in a $70 \times 150 \text{ mm}^2$ structure with 62% efficiency, whereas Rai *et al.* [9] reported a $150 \times 72.6 \text{ mm}^2$ antenna covering 3.2–7.3 GHz with 77% efficiency and $\text{ECC} < 0.069$.

This paper proposes a compact eight-element UWB MIMO antenna with an overall size of $70 \times 38 \text{ mm}^2$ for radar imaging and high-speed wireless applications. The antenna operates from 6.3 to 12.4 GHz with low mutual coupling, nearly omnidirectional radiation behavior, approximately 94% radiation efficiency, and a peak gain of about 4.9 dB. Section II describes the single-element and MIMO antenna designs. Section III presents the performance analysis in terms of reflection coefficient, isolation, ECC, diversity gain, CCL, TARC, and radiation characteristics. Sections IV and V present the comparison with related works and the conclusion, respectively.

II. ANTENNA DESIGN ANALYSIS

A. Single-Element Antenna Design Procedure

The single-element microstrip patch antenna is developed on a square substrate of $15 \times 15 \text{ mm}^2$, as shown in Fig. 1.

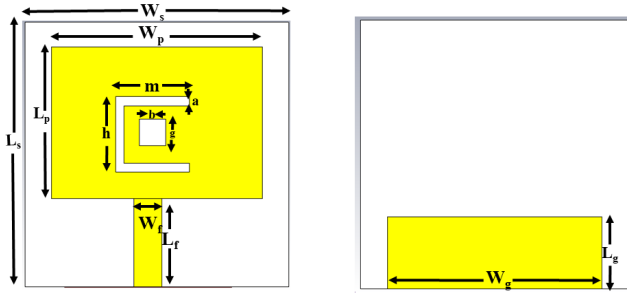


Fig. 1. Single-element antenna: (a) front view and (b) back view.

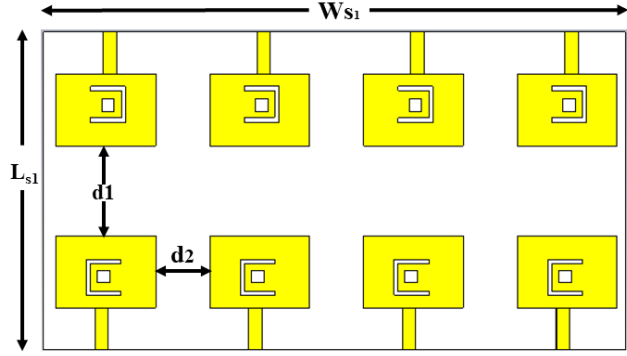


Fig. 2. Front view of the eight-element MIMO antenna.

The resonant frequency is tuned to the target operating band by placing a modified rectangular radiating patch on the top surface of the substrate. A microstrip feed line excites the radiator, while a modified slot structure etched on the patch increases the effective surface-current path and improves impedance matching. A partial ground plane with width 12 mm is used to improve bandwidth and radiation characteristics. The optimized parameters of the single-element antenna are listed in Table I.

TABLE I
PARAMETRIC STUDY OF SINGLE-ELEMENT ANTENNA

Symbol	Dimension (mm)	Symbol	Dimension (mm)
$W_s = L_s$	15	a	4.20
L_p	8.6	b	4.30
W_f	5.0	c	0.55
L_f	1.6	d	1.50
W_g	12	g	1.50

B. Eight-Element MIMO Antenna Design Procedure

The front and back views of the proposed eight-element MIMO antenna are shown in Figs. 2 and 3, respectively. Eight identical single elements are mounted on a common $70 \times 38 \text{ mm}^2$ substrate to realize spatial diversity and improved MIMO performance. The antenna elements are symmetrically placed along the top and bottom edges of the substrate to reduce mutual coupling and support pattern diversity. Each antenna element is fed independently through a microstrip feed line. The optimized MIMO antenna parameters are listed in Table II.

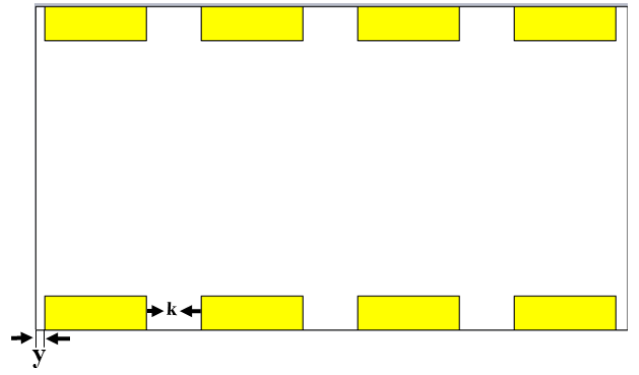


Fig. 3. Back view of the eight-element MIMO antenna.

TABLE II
PARAMETRIC STUDY OF MIMO ANTENNA SYSTEM

Symbol	Dimension (mm)	Symbol	Dimension (mm)
W_{ms}	70	d_1	6.50
L_{ms}	38	d_2	6.50
d_3	10.80	y	1.00

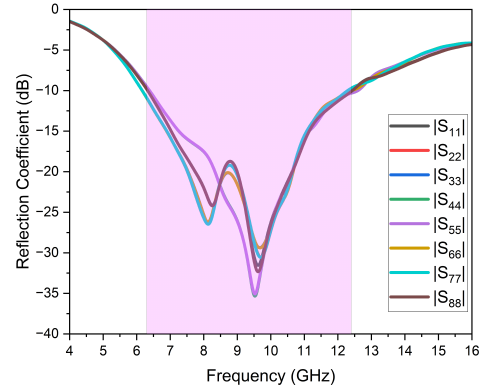


Fig. 4. Reflection coefficient of the proposed MIMO antenna.

III. PERFORMANCE ANALYSIS

Fig. 4 shows that the reflection coefficients of the eight ports remain below -10 dB over the $6.3\text{--}12.4 \text{ GHz}$ band, demonstrating good impedance matching across the UWB operating range. Two resonances are observed around 8.1 and 9.5 GHz , with minimum return losses of approximately -30 and -35 dB , respectively. The nearly identical responses of all ports indicate strong port-to-port consistency and structural symmetry, confirming the suitability of the proposed antenna for UWB radar sensing and high-speed wireless communication.

The port-isolation characteristics between antenna 1 and the remaining elements are shown in Fig. 5. The isolation is generally better than -11 dB and improves considerably at higher frequencies. Several transmission coefficients exhibit deep nulls below -35 dB , confirming effective electromagnetic decoupling among the antenna elements. These results indicate that the antenna array provides adequate isolation for

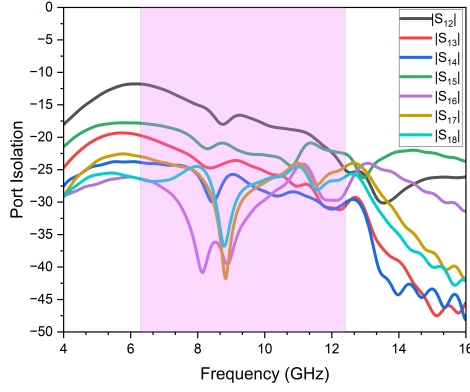


Fig. 5. Port isolation of the proposed MIMO antenna.

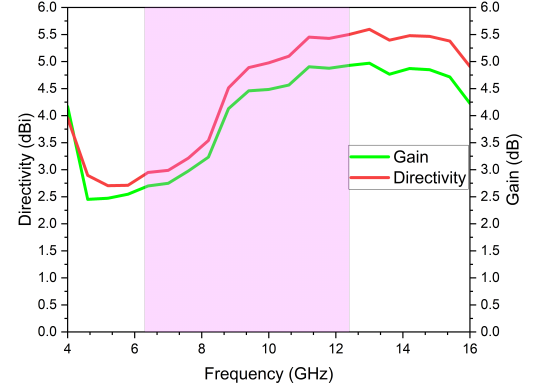


Fig. 7. Directivity versus gain analysis of the proposed MIMO antenna.

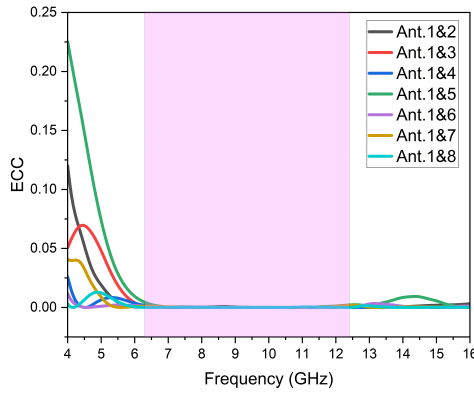


Fig. 6. ECC analysis of the proposed MIMO antenna.

practical MIMO operation across the considered UWB band.

Fig. 6 shows the envelope correlation coefficient between antenna 1 and the remaining antenna elements from 6.3 to 12.4 GHz. The ECC remains below 0.004 over the operating band, which is far lower than the common practical limit of 0.5 and also below the stricter 0.1 level for high-diversity systems. The near-zero ECC confirms weak correlation, good isolation, and improved channel capacity.

The gain and directivity responses of the proposed antenna are shown in Fig. 7. The antenna maintains stable radiation behavior across the operating band, with a peak directivity of about 5.4 dBi and a maximum gain of about 4.9 dB. The close agreement between gain and directivity indicates low radiation loss and efficient radiation performance.

The far-field 3-D directivity and gain radiation patterns at 9.6 GHz are shown in Fig. 8. The radiation patterns are stable and nearly symmetric, confirming well-distributed radiation within the target coverage region. The diversity gain is shown in Fig. 9; it remains close to the theoretical maximum of 10 dB and reaches approximately 9.97 dB around 6.3 GHz, verifying strong diversity performance and low correlation among antenna elements.

Fig. 10 illustrates the surface-current distribution of the

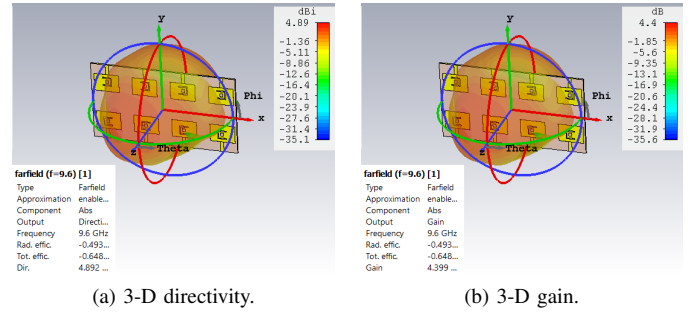


Fig. 8. (a) 3-D directivity and (b) 3-D gain of the proposed MIMO antenna.

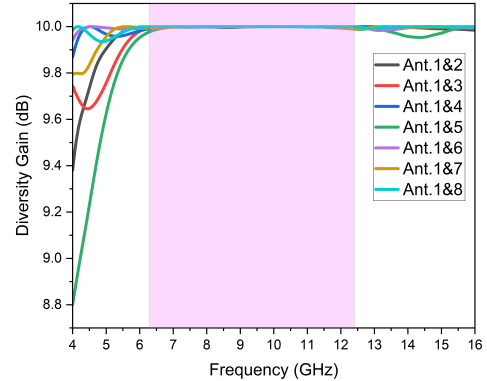


Fig. 9. Diversity gain of the proposed MIMO antenna.

antenna array at 9.6 GHz under $\phi = 0^\circ$. The maximum current concentration is approximately 98.1 A/m and is mainly observed near the feed lines and radiating edges. The current remains well confined to the excited elements, with limited leakage into adjacent elements, confirming low mutual coupling and supporting high MIMO efficiency.

Fig. 11 shows the radiation efficiency of the eight-element antenna system from 6.3 to 12.4 GHz. All antenna elements maintain high efficiency across the operating band, with an average value of about 94%. Peak efficiency above 95% is

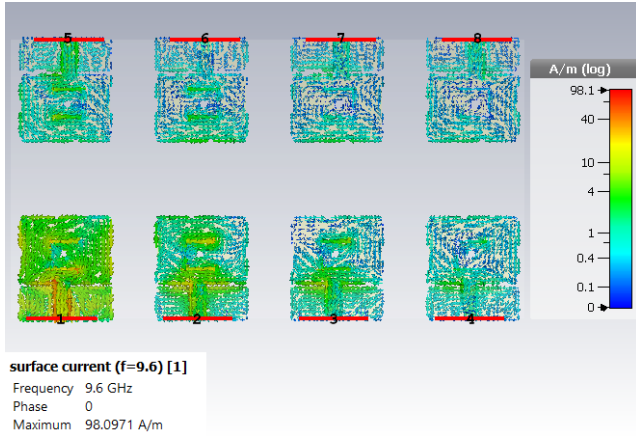


Fig. 10. Surface current distribution of the proposed MIMO antenna.

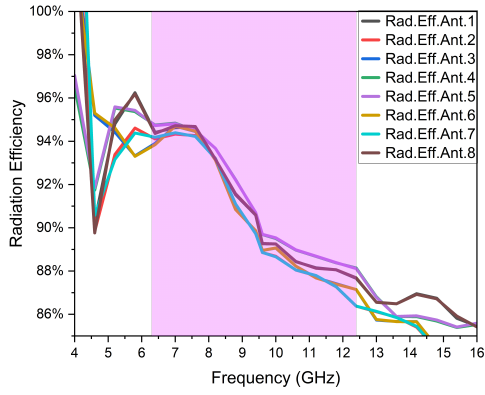


Fig. 11. Radiation efficiency of the proposed MIMO antenna.

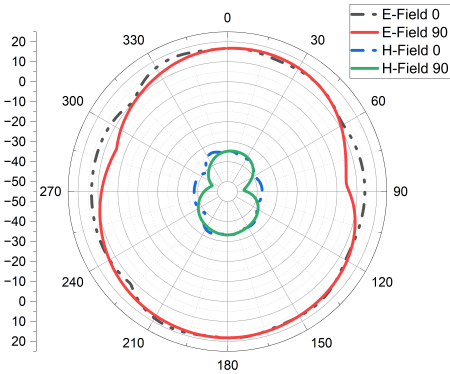


Fig. 12. Radiation pattern of the proposed MIMO antenna.

achieved in the mid-band, and the efficiency remains higher than 88% near the upper band edge.

The far-field radiation patterns of the proposed antenna are shown in Fig. 12. The E-field patterns in the 0° and 90° planes show nearly omnidirectional behavior in the azimuth plane, while the H-field patterns exhibit the conventional bidirectional response of planar antennas. These characteristics

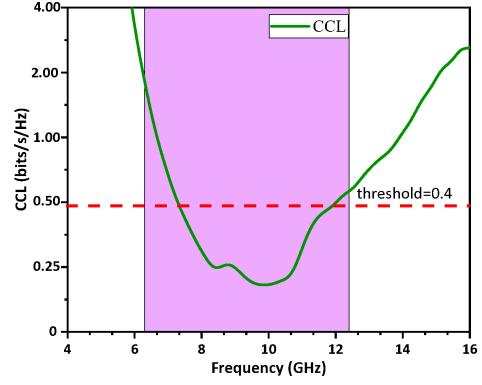


Fig. 13. CCL analysis of the proposed MIMO antenna.

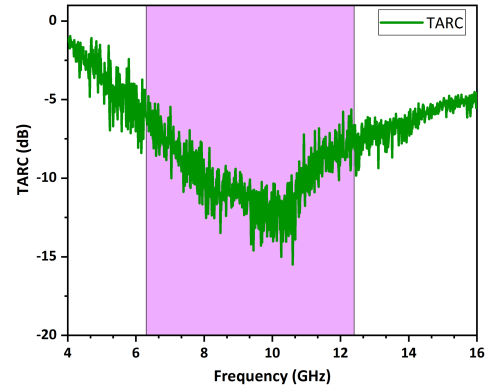


Fig. 14. TARC analysis of the proposed MIMO antenna.

support stable coverage for UWB MIMO wireless applications.

The channel capacity loss (CCL) and total active reflection coefficient (TARC) are presented in Figs. 13 and 14, respectively. The CCL remains below 0.4 bits/s/Hz across the operating band, confirming low channel-capacity degradation. The TARC remains close to or below -10 dB over a large portion of the UWB operating band, indicating stable multipoint operation and acceptable impedance matching under simultaneous port excitation.

IV. COMPARATIVE ANALYSIS

Table III compares the proposed antenna with recently reported designs [5]–[9]. The proposed antenna occupies only $70 \times 38 \text{ mm}^2$, which is considerably smaller than the structures reported in [5] and [8]. Although some earlier works report larger bandwidths or gains, they either use larger antenna sizes or do not provide complete MIMO metrics. The proposed design provides a balanced combination of compact size, 6.1 GHz bandwidth, 4.9 dB gain, 94% radiation efficiency, and ECC below 0.004, making it suitable for compact UWB radar imaging and short-range high-speed communication systems.

TABLE III
COMPARISON WITH RECENT RELEVANT WORK

Parameter	[5]	[6]	[7]	[8]	[9]	This work
Size (mm ²)	150 × 70	–	12 × 20	70 × 150	150 × 72.6	70 × 38
Freq. (GHz)	3.4–3.8, 10.5–14	2.75–12	3.1–10.6	6.425–8.4	3.2–7.3	6.3–12.4
BW (GHz)	9	14.9	9.25	2	4.1	6.1
Max. gain (dB)	5.65	9.65	3.4	–	–	4.9
Max. eff. (%)	61	–	89.3	62	77	94
ECC	< 0.15	–	< 0.18	–	< 0.069	< 0.004

V. CONCLUSION

This paper presented a compact eight-element UWB MIMO antenna for radar imaging and high-speed short-range wireless communication. The proposed antenna operates from 6.3 to 12.4 GHz, with two main resonances around 8.1 and 9.5 GHz, while maintaining stable impedance behavior across the operating band. The antenna achieves about 94% radiation efficiency and a peak gain of 4.9 dB. It also provides very low ECC below 0.004, strong isolation among elements, acceptable CCL, and acceptable TARC performance, confirming reliable multiport behavior under simultaneous excitation. Considering its compact size, wide bandwidth, high radiation efficiency, and favorable diversity characteristics, the antenna is suitable for UWB radar imaging, through-wall detection, indoor positioning, short-range vehicular radar, IoT-based high-speed data links, and multigigabit short-range wireless communication systems.

Future work may include experimental validation in practical UWB radar imaging scenarios and integration with compact wireless sensing platforms.

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